

Reinforcement Learning

Lab5

```
python -m pip install gymnasium

import gymnasium as gym

def random_agent(env_name):
    # 1. Create the environment
    env = gym.make(env_name)

    # 2. Reset the environment
    state, info = env.reset()

    total_reward = 0
    steps = 0

    print("\nEnvironment:", env_name)
    print("Initial State:", state)

    # 3-8. Run until episode terminates
    while True:

        # 4. Identify available actions
        # 5. Select an action randomly
        action = env.action_space.sample()

        # 6. Execute the action
        next_state, reward, terminated, truncated, info = env.step(action)

        # 7. Observe next state and reward
        print(
            "Step:", steps + 1,
```

```

        "| Action:", action,
        "| Reward:", reward,
        "| Next State:", next_state
    )

    total_reward += reward
    steps += 1
    state = next_state

    # 8. Check whether episode has terminated
    if terminated or truncated:
        break

    # 9. Calculate total and average reward
    average_reward = total_reward / steps

    print("\nTotal Reward:", total_reward)
    print("Average Reward:", average_reward)
    print("Total Steps:", steps)

    env.close()

# Test the three environments
random_agent("CartPole-v1")
random_agent("MountainCar-v0")
random_agent("FrozenLake-v1")

```

Output:

Environment: CartPole-v1

Initial State: [0.03064593 -0.044388 -0.00445641 0.00320612]

Step: 1 | Action: 0 | Reward: 1.0 | Next State: [0.02975817 -0.23944575 -0.00439228
0.29447967]

Step: 2 | Action: 1 | Reward: 1.0 | Next State: [0.02496926 -0.04426146 0.00149731
0.00041473]

Step: 3 | Action: 0 | Reward: 1.0 | Next State: [0.02408403 -0.23940486 0.00150561
0.29356968]

Step: 4 | Action: 1 | Reward: 1.0 | Next State: [0.01929593 -0.0443044 0.007377 0.001362]

Step: 5 | Action: 1 | Reward: 1.0 | Next State: [0.01840984 0.15071099 0.00740424
-0.2889843]
Step: 6 | Action: 1 | Reward: 1.0 | Next State: [0.02142406 0.34572658 0.00162455
-0.5793228]
Step: 7 | Action: 0 | Reward: 1.0 | Next State: [0.02833859 0.1505819 -0.0099619
-0.28612858]
Step: 8 | Action: 0 | Reward: 1.0 | Next State: [0.03135023 -0.04439658 -0.01568448
0.00339586]
Step: 9 | Action: 1 | Reward: 1.0 | Next State: [0.0304623 0.15094675 -0.01561656
-0.29419416]
Step: 10 | Action: 1 | Reward: 1.0 | Next State: [0.03348124 0.34628785 -0.02150044
-0.5917611]
Step: 11 | Action: 1 | Reward: 1.0 | Next State: [0.04040699 0.54170406 -0.03333566
-0.8911384]
Step: 12 | Action: 0 | Reward: 1.0 | Next State: [0.05124107 0.34704986 -0.05115843
-0.60911816]
Step: 13 | Action: 0 | Reward: 1.0 | Next State: [0.05818207 0.152679 -0.06334079
-0.3329773]
Step: 14 | Action: 0 | Reward: 1.0 | Next State: [0.06123565 -0.04148691 -0.07000034
-0.06092225]
Step: 15 | Action: 1 | Reward: 1.0 | Next State: [0.06040591 0.15456524 -0.07121879
-0.37484357]
Step: 16 | Action: 1 | Reward: 1.0 | Next State: [0.06349722 0.35062274 -0.07871566
-0.6891049]
Step: 17 | Action: 0 | Reward: 1.0 | Next State: [0.07050967 0.15667634 -0.09249776
-0.42220467]
Step: 18 | Action: 1 | Reward: 1.0 | Next State: [0.0736432 0.35297868 -0.10094185
-0.742555]
Step: 19 | Action: 1 | Reward: 1.0 | Next State: [0.08070277 0.54933834 -0.11579295
-1.0652218]
Step: 20 | Action: 1 | Reward: 1.0 | Next State: [0.09168954 0.74578637 -0.13709739
-1.3918877]
Step: 21 | Action: 1 | Reward: 1.0 | Next State: [0.10660527 0.942323 -0.16493514
-1.7241069]
Step: 22 | Action: 1 | Reward: 1.0 | Next State: [0.12545173 1.1389039 -0.19941728
-2.0632477]
Step: 23 | Action: 0 | Reward: 1.0 | Next State: [0.14822981 0.94629693 -0.24068223
-1.8383038]

Total Reward: 23.0

Average Reward: 1.0

Total Steps: 23

Environment: MountainCar-v0

Initial State: [-0.44897622 0.]

Step: 1 | Action: 2 | Reward: -1.0 | Next State: [-4.4853121e-01 4.4499384e-04]
Step: 2 | Action: 1 | Reward: -1.0 | Next State: [-4.48644489e-01 -1.13265975e-04]
Step: 3 | Action: 0 | Reward: -1.0 | Next State: [-0.45031518 -0.0016707]
Step: 4 | Action: 2 | Reward: -1.0 | Next State: [-0.45153108 -0.00121591]
Step: 5 | Action: 0 | Reward: -1.0 | Next State: [-0.4542833 -0.00275221]
Step: 6 | Action: 1 | Reward: -1.0 | Next State: [-0.45755163 -0.00326834]
Step: 7 | Action: 0 | Reward: -1.0 | Next State: [-0.4623121 -0.00476046]
Step: 8 | Action: 0 | Reward: -1.0 | Next State: [-0.46852964 -0.00621753]
Step: 9 | Action: 2 | Reward: -1.0 | Next State: [-0.4741583 -0.00562867]
Step: 10 | Action: 1 | Reward: -1.0 | Next State: [-0.48015642 -0.00599811]
Step: 11 | Action: 0 | Reward: -1.0 | Next State: [-0.48747942 -0.00732301]
Step: 12 | Action: 1 | Reward: -1.0 | Next State: [-0.49507278 -0.00759337]
Step: 13 | Action: 2 | Reward: -1.0 | Next State: [-0.5018799 -0.00680706]
Step: 14 | Action: 2 | Reward: -1.0 | Next State: [-0.5078497 -0.00596983]
Step: 15 | Action: 1 | Reward: -1.0 | Next State: [-0.5139376 -0.00608791]
Step: 16 | Action: 0 | Reward: -1.0 | Next State: [-0.52109796 -0.00716036]
Step: 17 | Action: 0 | Reward: -1.0 | Next State: [-0.5292771 -0.00817911]
Step: 18 | Action: 1 | Reward: -1.0 | Next State: [-0.5374136 -0.00813653]
Step: 19 | Action: 1 | Reward: -1.0 | Next State: [-0.5454465 -0.00803295]
Step: 20 | Action: 1 | Reward: -1.0 | Next State: [-0.55331576 -0.00786921]
Step: 21 | Action: 2 | Reward: -1.0 | Next State: [-0.5599624 -0.00664662]
Step: 22 | Action: 1 | Reward: -1.0 | Next State: [-0.5663368 -0.00637444]
Step: 23 | Action: 0 | Reward: -1.0 | Next State: [-0.5733916 -0.00705478]
Step: 24 | Action: 0 | Reward: -1.0 | Next State: [-0.5810743 -0.00768272]
Step: 25 | Action: 1 | Reward: -1.0 | Next State: [-0.5883281 -0.00725379]
Step: 26 | Action: 2 | Reward: -1.0 | Next State: [-0.59409946 -0.00577136]
Step: 27 | Action: 2 | Reward: -1.0 | Next State: [-0.598346 -0.00424654]
Step: 28 | Action: 1 | Reward: -1.0 | Next State: [-0.60203665 -0.00369062]
Step: 29 | Action: 1 | Reward: -1.0 | Next State: [-0.6051444 -0.00310775]
Step: 30 | Action: 0 | Reward: -1.0 | Next State: [-0.60864663 -0.00350224]
Step: 31 | Action: 2 | Reward: -1.0 | Next State: [-0.6105179 -0.00187128]
Step: 32 | Action: 2 | Reward: -1.0 | Next State: [-6.1074466e-01 -2.2675177e-04]
Step: 33 | Action: 2 | Reward: -1.0 | Next State: [-0.60932523 0.00141942]
Step: 34 | Action: 2 | Reward: -1.0 | Next State: [-0.60626996 0.00305531]
Step: 35 | Action: 0 | Reward: -1.0 | Next State: [-0.6036009 0.002669]
Step: 36 | Action: 0 | Reward: -1.0 | Next State: [-0.6013377 0.00226328]
Step: 37 | Action: 0 | Reward: -1.0 | Next State: [-0.5994966 0.00184105]
Step: 38 | Action: 0 | Reward: -1.0 | Next State: [-0.59809124 0.00140537]
Step: 39 | Action: 0 | Reward: -1.0 | Next State: [-0.5971318 0.00095943]
Step: 40 | Action: 1 | Reward: -1.0 | Next State: [-0.59562534 0.00150646]
Step: 41 | Action: 1 | Reward: -1.0 | Next State: [-0.59358287 0.00204247]
Step: 42 | Action: 1 | Reward: -1.0 | Next State: [-0.5910194 0.0025635]

Step: 43 | Action: 0 | Reward: -1.0 | Next State: [-0.5889537 0.00206572]
Step: 44 | Action: 2 | Reward: -1.0 | Next State: [-0.5854009 0.00355274]
Step: 45 | Action: 2 | Reward: -1.0 | Next State: [-0.5803873 0.00501361]
Step: 46 | Action: 1 | Reward: -1.0 | Next State: [-0.57494986 0.00543747]
Step: 47 | Action: 1 | Reward: -1.0 | Next State: [-0.56912875 0.00582108]
Step: 48 | Action: 1 | Reward: -1.0 | Next State: [-0.56296724 0.00616149]
Step: 49 | Action: 1 | Reward: -1.0 | Next State: [-0.5565112 0.00645607]
Step: 50 | Action: 1 | Reward: -1.0 | Next State: [-0.5498087 0.00670251]
Step: 51 | Action: 1 | Reward: -1.0 | Next State: [-0.5429098 0.00689888]
Step: 52 | Action: 1 | Reward: -1.0 | Next State: [-0.53586614 0.00704363]
Step: 53 | Action: 1 | Reward: -1.0 | Next State: [-0.5287306 0.00713562]
Step: 54 | Action: 2 | Reward: -1.0 | Next State: [-0.52055645 0.00817411]
Step: 55 | Action: 1 | Reward: -1.0 | Next State: [-0.51240516 0.00815129]
Step: 56 | Action: 2 | Reward: -1.0 | Next State: [-0.5033378 0.00906735]
Step: 57 | Action: 0 | Reward: -1.0 | Next State: [-0.4954223 0.00791549]
Step: 58 | Action: 2 | Reward: -1.0 | Next State: [-0.4867179 0.00870442]
Step: 59 | Action: 1 | Reward: -1.0 | Next State: [-0.4782895 0.00842837]
Step: 60 | Action: 0 | Reward: -1.0 | Next State: [-0.47119993 0.0070896]
Step: 61 | Action: 2 | Reward: -1.0 | Next State: [-0.4635017 0.00769823]
Step: 62 | Action: 1 | Reward: -1.0 | Next State: [-0.45625177 0.00724994]
Step: 63 | Action: 0 | Reward: -1.0 | Next State: [-0.4505035 0.00574826]
Step: 64 | Action: 1 | Reward: -1.0 | Next State: [-0.44529906 0.00520443]
Step: 65 | Action: 1 | Reward: -1.0 | Next State: [-0.4406765 0.00462257]
Step: 66 | Action: 0 | Reward: -1.0 | Next State: [-0.43766946 0.00300705]
Step: 67 | Action: 1 | Reward: -1.0 | Next State: [-0.43529975 0.00236969]
Step: 68 | Action: 2 | Reward: -1.0 | Next State: [-0.4325846 0.00271517]
Step: 69 | Action: 0 | Reward: -1.0 | Next State: [-0.4315436 0.00104101]
Step: 70 | Action: 2 | Reward: -1.0 | Next State: [-0.43018425 0.00135934]
Step: 71 | Action: 2 | Reward: -1.0 | Next State: [-0.4285164 0.00166786]
Step: 72 | Action: 2 | Reward: -1.0 | Next State: [-0.42655203 0.00196438]
Step: 73 | Action: 1 | Reward: -1.0 | Next State: [-0.42530525 0.00124677]
Step: 74 | Action: 1 | Reward: -1.0 | Next State: [-0.42478505 0.0005202]
Step: 75 | Action: 1 | Reward: -1.0 | Next State: [-4.2499515e-01 -2.1009461e-04]
Step: 76 | Action: 1 | Reward: -1.0 | Next State: [-0.42593402 -0.00093888]
Step: 77 | Action: 0 | Reward: -1.0 | Next State: [-0.42859495 -0.00266093]
Step: 78 | Action: 0 | Reward: -1.0 | Next State: [-0.4329588 -0.00436386]
Step: 79 | Action: 0 | Reward: -1.0 | Next State: [-0.4389941 -0.00603531]
Step: 80 | Action: 2 | Reward: -1.0 | Next State: [-0.44465718 -0.00566305]
Step: 81 | Action: 2 | Reward: -1.0 | Next State: [-0.44990677 -0.0052496]
Step: 82 | Action: 2 | Reward: -1.0 | Next State: [-0.45470455 -0.00479779]
Step: 83 | Action: 0 | Reward: -1.0 | Next State: [-0.4610154 -0.00631083]
Step: 84 | Action: 1 | Reward: -1.0 | Next State: [-0.46779284 -0.00677745]
Step: 85 | Action: 2 | Reward: -1.0 | Next State: [-0.4739869 -0.00619404]
Step: 86 | Action: 2 | Reward: -1.0 | Next State: [-0.47955164 -0.00556476]

Step: 87 | Action: 2 | Reward: -1.0 | Next State: [-0.4844458 -0.00489415]
Step: 88 | Action: 1 | Reward: -1.0 | Next State: [-0.48963293 -0.00518713]
Step: 89 | Action: 2 | Reward: -1.0 | Next State: [-0.49407437 -0.00444143]
Step: 90 | Action: 0 | Reward: -1.0 | Next State: [-0.49973693 -0.00566257]
Step: 91 | Action: 2 | Reward: -1.0 | Next State: [-0.50457835 -0.00484138]
Step: 92 | Action: 0 | Reward: -1.0 | Next State: [-0.5105623 -0.00598396]
Step: 93 | Action: 1 | Reward: -1.0 | Next State: [-0.516644 -0.00608171]
Step: 94 | Action: 2 | Reward: -1.0 | Next State: [-0.52177787 -0.00513387]
Step: 95 | Action: 0 | Reward: -1.0 | Next State: [-0.5279254 -0.00614752]
Step: 96 | Action: 0 | Reward: -1.0 | Next State: [-0.53504044 -0.00711507]
Step: 97 | Action: 1 | Reward: -1.0 | Next State: [-0.54206973 -0.00702928]
Step: 98 | Action: 1 | Reward: -1.0 | Next State: [-0.54896057 -0.00689082]
Step: 99 | Action: 1 | Reward: -1.0 | Next State: [-0.5556613 -0.00670079]
Step: 100 | Action: 2 | Reward: -1.0 | Next State: [-0.56112206 -0.00546069]
Step: 101 | Action: 1 | Reward: -1.0 | Next State: [-0.5663019 -0.00517986]
Step: 102 | Action: 2 | Reward: -1.0 | Next State: [-0.57016236 -0.00386046]
Step: 103 | Action: 1 | Reward: -1.0 | Next State: [-0.57367474 -0.00351237]
Step: 104 | Action: 0 | Reward: -1.0 | Next State: [-0.5778129 -0.00413821]
Step: 105 | Action: 1 | Reward: -1.0 | Next State: [-0.5815463 -0.00373339]
Step: 106 | Action: 1 | Reward: -1.0 | Next State: [-0.5848473 -0.00330097]
Step: 107 | Action: 1 | Reward: -1.0 | Next State: [-0.5876915 -0.00284419]
Step: 108 | Action: 1 | Reward: -1.0 | Next State: [-0.5900579 -0.00236645]
Step: 109 | Action: 2 | Reward: -1.0 | Next State: [-0.59092927 -0.0008713]
Step: 110 | Action: 2 | Reward: -1.0 | Next State: [-0.590299 0.00063025]
Step: 111 | Action: 1 | Reward: -1.0 | Next State: [-0.5891718 0.00112717]
Step: 112 | Action: 0 | Reward: -1.0 | Next State: [-0.58855605 0.0006158]
Step: 113 | Action: 2 | Reward: -1.0 | Next State: [-0.5864561 0.0020999]
Step: 114 | Action: 0 | Reward: -1.0 | Next State: [-0.58488756 0.00156854]
Step: 115 | Action: 2 | Reward: -1.0 | Next State: [-0.581862 0.00302562]
Step: 116 | Action: 1 | Reward: -1.0 | Next State: [-0.57840157 0.00346038]
Step: 117 | Action: 0 | Reward: -1.0 | Next State: [-0.575532 0.00286955]
Step: 118 | Action: 2 | Reward: -1.0 | Next State: [-0.5712746 0.00425747]
Step: 119 | Action: 0 | Reward: -1.0 | Next State: [-0.56766075 0.00361382]
Step: 120 | Action: 0 | Reward: -1.0 | Next State: [-0.5647174 0.00294333]
Step: 121 | Action: 0 | Reward: -1.0 | Next State: [-0.56246644 0.00225094]
Step: 122 | Action: 2 | Reward: -1.0 | Next State: [-0.5589247 0.00354178]
Step: 123 | Action: 1 | Reward: -1.0 | Next State: [-0.55511844 0.00380623]
Step: 124 | Action: 1 | Reward: -1.0 | Next State: [-0.5510762 0.00404228]
Step: 125 | Action: 2 | Reward: -1.0 | Next State: [-0.54582804 0.00524812]
Step: 126 | Action: 2 | Reward: -1.0 | Next State: [-0.53941333 0.00641472]
Step: 127 | Action: 0 | Reward: -1.0 | Next State: [-0.53388005 0.00553328]
Step: 128 | Action: 1 | Reward: -1.0 | Next State: [-0.52826965 0.00561038]
Step: 129 | Action: 2 | Reward: -1.0 | Next State: [-0.52162427 0.00664541]
Step: 130 | Action: 2 | Reward: -1.0 | Next State: [-0.5139937 0.0076306]

Step: 131 | Action: 1 | Reward: -1.0 | Next State: [-0.5064351 0.00755858]
Step: 132 | Action: 0 | Reward: -1.0 | Next State: [-0.5000052 0.0064299]
Step: 133 | Action: 2 | Reward: -1.0 | Next State: [-0.49275208 0.0072531]
Step: 134 | Action: 1 | Reward: -1.0 | Next State: [-0.48573 0.00702208]
Step: 135 | Action: 0 | Reward: -1.0 | Next State: [-0.47999132 0.00573867]
Step: 136 | Action: 2 | Reward: -1.0 | Next State: [-0.47357875 0.00641255]
Step: 137 | Action: 2 | Reward: -1.0 | Next State: [-0.46653995 0.00703881]
Step: 138 | Action: 0 | Reward: -1.0 | Next State: [-0.460927 0.00561295]
Step: 139 | Action: 0 | Reward: -1.0 | Next State: [-0.45678133 0.00414568]
Step: 140 | Action: 1 | Reward: -1.0 | Next State: [-0.45313343 0.0036479]
Step: 141 | Action: 2 | Reward: -1.0 | Next State: [-0.4490101 0.00412333]
Step: 142 | Action: 1 | Reward: -1.0 | Next State: [-0.4454415 0.00356858]
Step: 143 | Action: 2 | Reward: -1.0 | Next State: [-0.44145375 0.00398775]
Step: 144 | Action: 2 | Reward: -1.0 | Next State: [-0.43707588 0.00437788]
Step: 145 | Action: 1 | Reward: -1.0 | Next State: [-0.43333966 0.00373622]
Step: 146 | Action: 2 | Reward: -1.0 | Next State: [-0.42927215 0.00406752]
Step: 147 | Action: 1 | Reward: -1.0 | Next State: [-0.42590266 0.00336948]
Step: 148 | Action: 2 | Reward: -1.0 | Next State: [-0.42225546 0.0036472]
Step: 149 | Action: 2 | Reward: -1.0 | Next State: [-0.4183567 0.00389878]
Step: 150 | Action: 0 | Reward: -1.0 | Next State: [-0.41623417 0.00212251]
Step: 151 | Action: 0 | Reward: -1.0 | Next State: [-4.1590303e-01 3.3113040e-04]
Step: 152 | Action: 1 | Reward: -1.0 | Next State: [-0.41636565 -0.00046261]
Step: 153 | Action: 2 | Reward: -1.0 | Next State: [-4.1661870e-01 -2.5305658e-04]
Step: 154 | Action: 0 | Reward: -1.0 | Next State: [-0.4186604 -0.0020417]
Step: 155 | Action: 1 | Reward: -1.0 | Next State: [-0.42147622 -0.00281581]
Step: 156 | Action: 2 | Reward: -1.0 | Next State: [-0.424046 -0.0025698]
Step: 157 | Action: 2 | Reward: -1.0 | Next State: [-0.4263514 -0.00230539]
Step: 158 | Action: 1 | Reward: -1.0 | Next State: [-0.42937586 -0.00302445]
Step: 159 | Action: 1 | Reward: -1.0 | Next State: [-0.4330976 -0.00372175]
Step: 160 | Action: 2 | Reward: -1.0 | Next State: [-0.4364898 -0.0033922]
Step: 161 | Action: 1 | Reward: -1.0 | Next State: [-0.4405279 -0.0040381]
Step: 162 | Action: 0 | Reward: -1.0 | Next State: [-0.4461826 -0.00565471]
Step: 163 | Action: 1 | Reward: -1.0 | Next State: [-0.45241272 -0.00623012]
Step: 164 | Action: 2 | Reward: -1.0 | Next State: [-0.45817268 -0.00575997]
Step: 165 | Action: 1 | Reward: -1.0 | Next State: [-0.4644202 -0.00624752]
Step: 166 | Action: 0 | Reward: -1.0 | Next State: [-0.47210923 -0.00768903]
Step: 167 | Action: 0 | Reward: -1.0 | Next State: [-0.4811829 -0.00907367]
Step: 168 | Action: 0 | Reward: -1.0 | Next State: [-0.49157384 -0.01039093]
Step: 169 | Action: 0 | Reward: -1.0 | Next State: [-0.5032046 -0.01163075]
Step: 170 | Action: 0 | Reward: -1.0 | Next State: [-0.5159882 -0.01278361]
Step: 171 | Action: 0 | Reward: -1.0 | Next State: [-0.5298289 -0.01384068]
Step: 172 | Action: 2 | Reward: -1.0 | Next State: [-0.54262286 -0.01279396]
Step: 173 | Action: 1 | Reward: -1.0 | Next State: [-0.5552742 -0.01265136]
Step: 174 | Action: 2 | Reward: -1.0 | Next State: [-0.56668836 -0.01141415]

Step: 175 | Action: 2 | Reward: -1.0 | Next State: [-0.5767802 -0.01009188]
Step: 176 | Action: 1 | Reward: -1.0 | Next State: [-0.5864749 -0.0096947]
Step: 177 | Action: 2 | Reward: -1.0 | Next State: [-0.5947009 -0.00822593]
Step: 178 | Action: 0 | Reward: -1.0 | Next State: [-0.60339755 -0.00869669]
Step: 179 | Action: 1 | Reward: -1.0 | Next State: [-0.61150146 -0.0081039]
Step: 180 | Action: 0 | Reward: -1.0 | Next State: [-0.6199537 -0.00845225]
Step: 181 | Action: 0 | Reward: -1.0 | Next State: [-0.6286933 -0.00873961]
Step: 182 | Action: 0 | Reward: -1.0 | Next State: [-0.6376577 -0.00896439]
Step: 183 | Action: 2 | Reward: -1.0 | Next State: [-0.64478326 -0.00712555]
Step: 184 | Action: 0 | Reward: -1.0 | Next State: [-0.6520198 -0.00723655]
Step: 185 | Action: 2 | Reward: -1.0 | Next State: [-0.65731686 -0.00529705]
Step: 186 | Action: 0 | Reward: -1.0 | Next State: [-0.6626377 -0.00532084]
Step: 187 | Action: 1 | Reward: -1.0 | Next State: [-0.6669457 -0.00430803]
Step: 188 | Action: 0 | Reward: -1.0 | Next State: [-0.6712115 -0.00426576]
Step: 189 | Action: 1 | Reward: -1.0 | Next State: [-0.674406 -0.00319449]
Step: 190 | Action: 1 | Reward: -1.0 | Next State: [-0.6765076 -0.00210163]
Step: 191 | Action: 2 | Reward: -1.0 | Next State: [-6.765022e-01 5.383930e-06]
Step: 192 | Action: 0 | Reward: -1.0 | Next State: [-6.76389873e-01 1.12364214e-04]
Step: 193 | Action: 2 | Reward: -1.0 | Next State: [-0.67417127 0.00221859]
Step: 194 | Action: 1 | Reward: -1.0 | Next State: [-0.6708614 0.00330987]
Step: 195 | Action: 1 | Reward: -1.0 | Next State: [-0.6664826 0.00437876]
Step: 196 | Action: 2 | Reward: -1.0 | Next State: [-0.66006476 0.00641787]
Step: 197 | Action: 0 | Reward: -1.0 | Next State: [-0.6536518 0.00641301]
Step: 198 | Action: 0 | Reward: -1.0 | Next State: [-0.6472879 0.00636385]
Step: 199 | Action: 1 | Reward: -1.0 | Next State: [-0.6400175 0.00727038]
Step: 200 | Action: 0 | Reward: -1.0 | Next State: [-0.63289165 0.00712587]

Total Reward: -200.0
Average Reward: -1.0
Total Steps: 200

Environment: FrozenLake-v1

Initial State: 0

Step: 1 | Action: 3 | Reward: 0 | Next State: 0

Step: 2 | Action: 2 | Reward: 0 | Next State: 1

Step: 3 | Action: 2 | Reward: 0 | Next State: 5

Total Reward: 0
Average Reward: 0.0
Total Steps: 3